Little Tiger Invitational - Repaired practice edition - Answer key

Division B | Thermodynamics | Repair appendix

This appendix contains authored replacements and supplied written alternatives. It does not reconstruct unpublished historical material. For the listed item IDs, use this appendix instead of the archived wording and key.

ANSWER KEY - Review only after submitting.

24. (1 points) Authored replacement for source item 24: For a closed system performing only pressure-volume boundary work, which process has zero boundary work because its volume is constant?

answer: Isochoric. Boundary work is the integral of pressure with respect to volume; dV=0. This does not rule out other forms of work. (1 points).

30. (1 points) Authored replacement for source item 30: State the Kelvin-Planck form of the second law in your own words, without attributing it to the wrong scientist.

answer: A cyclic heat engine cannot have as its sole effect extracting heat from a single reservoir and converting it entirely to work; some other effect, such as heat rejection, is required. (1 points).

32. (3 points) Authored replacement for source item 32: Calculate the energy required to heat 16.50 kg of silver from 20 C to its melting point and melt it completely. Added assumption: melting point is 962 C. Use c=230 J/(kg K), latent heat of fusion 88,000 J/kg, and no heat loss.

answer: 5026890 J (3 points).

Numerical target 5026890; absolute tolerance 10053.78. Accepted units: , J.

repair-case. (4 points) Authored written substitute for unavailable source material: Why check delta U from temperature too?

Unit: Internal energy and the first law.
Worked-case data: A system receives 500 J heat and does 200 J work. Find delta U and chemistry work.

answer: It independently verifies transfer accounting under the ideal model. (4 points).